

Reducing False-Positive PASS Verdicts from an LLM Evaluator

A small empirical study on LLM-as-a-judge reliability

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Context

This is a work sample distilled from a production multi-agent LLM system I built and operate, which uses LLMs to evaluate candidates against strict, disqualifying requirements for public tenders. In that setting the dangerous error is the **false positive** — passing a candidate who should be rejected. This study measures three ways to reduce that error, across two models.

Research question

When an LLM judges whether a candidate meets a requirement, does (a) an evidence-grounded prompt, and (b) a cheap deterministic verifier placed *in front of* the model, reduce the harmful-direction (false-positive) error — and does a stronger model need the scaffolding less?

Method

- **Dataset:** 40 synthetic CVs (25 ineligible, 15 eligible), written without giveaways, spanning realistic traps: in-progress / undated / specialization-not-master degrees, wrong-field master, wrong base profession, no master, insufficient experience.
- **Conditions:** (1) naive prompt; (2) evidence-grounded prompt (absence of evidence → reject); (3) grounded + a rule-based verifier that can only tighten an 'eligible' verdict ('rule first, LLM last').
- **Models:** gpt-4o-mini and gpt-4o, temperature 0, 5 runs/candidate, majority vote.
- **Metrics:** accuracy; false-positive rate with a Wilson 95% CI; false-negative rate; cost; latency.

Results

Model	Condition	Acc.	FPR	FPR 95% CI	FN
gpt-4o-mini (\$0.024)	naive	0.925	0.12	[0.04, 0.30]	0
	grounded	0.925	0.12	[0.04, 0.30]	0
	grounded + verifier	1.000	0.00	[0.00, 0.13]	0
gpt-4o (\$0.41)	naive	0.950	0.08	[0.02, 0.25]	0
	grounded	0.975	0.04	[0.01, 0.20]	0
	grounded + verifier	1.000	0.00	[0.00, 0.13]	0

Findings

- **The stronger model has a lower baseline error** (gpt-4o 0.08 vs gpt-4o-mini 0.12 FPR).
- **Prompt grounding helped only the stronger model, and modestly** (gpt-4o 0.08→0.04; gpt-4o-mini unchanged). A prior n=18 pilot overstated this effect — expanding the eval changed the conclusion, which is itself the point of doing evals properly.

- **A free deterministic verifier drove FPR to 0 for both models**, with no false negatives.
- **Cheaper and better than scaling the model**: gpt-4o-mini + verifier (FPR 0, \$0.024) beat gpt-4o alone (FPR 0.04, \$0.41 — ~17× the cost).
- **Wide confidence intervals** (n=25 ineligible): the point differences between prompts should not be over-read; the robust, separable result is the verifier reaching 0.

Honest caveat & takeaway

The verifier is perfect here *because every disqualifier in this dataset is expressible as a rule* — by construction. In a real corpus the hard cases are exactly those that are not cleanly rule-expressible, which is why the production system keeps a rule-based layer for the ~99% it can decide and reserves the LLM for the residual. The generalizable lesson: don't assume an LLM judge is reliable — measure the error that actually hurts, and prefer a deterministic guard wherever the decision is expressible as one, reserving the model (and the eval budget) for the fraction rules genuinely cannot decide.

Reproducible code, dataset, and full write-up: github.com/jorgegarcia205/betagroup-ai-army/tree/main/evals